# Neural Bellman Operators
# Author Response to the Advisory Referee

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## Author revision, original research identity and evidence boundary

This is an author revision of Neural Bellman Operators for Econometrica numerical methods. The controlling persisted advisory is R97, report commit `9723c6cd2cc8b3e071422f3e522d4a843946db63`, reviewing manuscript `f802fe64d91d9f5c1c8f86123b0fbe114e622279`. It is an owner-commissioned AI advisory, not an Econometric Society decision or a review of the R122 study. We retain the original recursive economic objective, continuation critics, feasible actors and research goal. The main new result turns a paid original Bellman support into exact action information and a checked policy replacement. Every conventional representation receives the same calculation.

The completed R120 author source and publication remain the reliable complete manuscript baseline. The actual R121 actors and pending native independent handoff have their separate immutable identities; their preserved results do not replace the newly acquired R122 sample. The refreshed complete 363-head census changes only the R121 head from its earlier state freeze to its actual primary commit and adds the current R122 branch. All 35 original report bodies and independently constrained historical episodes retain their source identities and previous response routes.

An earlier unpublished R122 response was lost during workspace interruption. The subsequently published recovery response and its generator are preserved byte for byte in a separate before-state archive. The present revision integrates the actual completed recovery and current proof review as new author source; it does not claim recovery of unpublished lost bytes. Every original R97 request and each entire R121 response subsection below are recovered from authenticated original files. Prior labels are normalized to R122 and prior execution/remaining-requirement headings are dated as retained R121 accounts; the substantive prior paragraphs are unchanged. Current R122 action and remaining requirements are appended separately. The machine map nests every original R121 item record unchanged and binds current source and actual saved evidence.

A new analytic deduction, Proposition S.22, was derived after F and is separately source-bound. It gives matching deterministic/randomized guarantees for the exact paid-support information interface, including the bounded implemented-future discrepancy. Its complete proof preserves the distinction between an abstract information class and the realized investment model or finite numerical service. It changes no sample, algorithm, protocol, numerical outcome or inference family.

Theorem S.24 and Propositions S.25–S.26 add a newly reconstructed, independently reviewed original-objective gain recursion. The finite-horizon proof, complete dual-menu minimum flow and discounted bounded-measurable version specify all-state incumbent and risk-dual premises. The invariant domain covers every admitted original or conservative lower-map kernel; the finite

occupation calculation requires a nonempty complete menu with finite real penalties. This analytic extension does not claim an executed longer-horizon tournament or a neural-specific convergence theorem.

Scientific source F, state source D, actual actors M, independent schema repair I122 and the reporting repair have distinct published commits. No frozen scientific bytes, sample, policy, parameter, inference family or original objective have been changed. The independent schema repair maps a numeric temporary alias after exact checks. The reporting repair also restores the pre-freeze matching-first-16 economic rule for timed price tables; it is a disclosed protocol-conformance correction, and all original frozen-code diagnostic rows remain preserved. Failed and incomplete attempts are retained with their observed costs and unknown bounds.

**Actual current execution scope.** The executed complete report retains 5,568 primary decisions and 8,352 timing decisions, with 174 policy cells, 116 own-control contrasts and 348 signed comparisons. The interventions change 824 actions; direct independent original primitives confirm 824 of those accepted gain claims and contain 824 accepted candidate intervals. The safe-gain population family has 102 strictly positive lower endpoints; signed comparisons have 0 strictly positive and 0 strictly negative population intervals. Full-menu sufficient ReLU regions are nonempty in 0 of 96 primary scenarios, 0 of 288 individual timing scenarios and 0 of 96 three-clock intersections. These counts distinguish proof-supported safe gains, direct independent confirmation and conditional sample/clock price domains. The original independent and reporter failures remain preserved; successful execution belongs to their explicitly published adapters. Detailed raw intervals, unresolved checks and upper not-ruled-out domains remain in the complete saved tables. The executed restrictive saved fourth-clock calculation adds 7,200 comparator constraints and retains 96 full-menu domains; 0 sufficient and 0 upper not-ruled-out domains are nonempty. Every four-clock domain is checked to be a subset of the original three-clock domain. This is an additional saved-clock restriction, with no retiming or new scientific observation. An earlier unpublished completion was observed before a runtime reset but its native suffix and final reporting bytes were not recovered from that durable checkpoint. The current completed recovery has its own explicit audit/reporting paths and saved native Git transports, with unchanged F/D/M/I/R identities and no new state, actor, service clock or selection rule. A subsequent large-upload interruption lost unpublished administrative, delivery and local diagnostic files; it did not erase the completed recovery evidence already published in Git. Those administrative sources are newly reconstructed with new identities. No lost-object hash is substitute evidence, and none of the unknown lost work is zero cost; the full research total remains unknown.

**How to read this response.** The complete preceding R120/R121 treatment under each request is historical, with its own experiment and source. The appended R122 paragraphs identify the present mathematical or algorithmic modification and the remaining scientific obligation. A completed calculation is not a blanket closure of all requests. All primary sources, quoted-request hashes, prior maps, failed attempts and actual result receipts are listed in `BASELINE122.json`, `CONTENT_RETENTION_MAP122.md` and `provenance122/R97_REQUIREMENTS122.json`.

## 0.1 B1. Neural specificity remains unestablished

**Original advisory request.**

> Validity, recursive qualification, the AVaR enclosure, the information lower bound, and the fallback are representation-neutral. The ReLU family is not the complete-work

winner and is never certified adopted over quadratic least squares.

**Completed R120 author response.** The completed study gives every representation the same four services and original-model information. The ordinary hinge receives the ReLU features and coefficients and pays the full learned-feature acquisition charge. All 512 ReLU–hinge action pairs are exactly equal. All 42 learned own-incumbent cells have strictly positive independent mean-gain lower bounds, but every one of the 14 learned completed cells has a strictly worse sample mean original objective than the online semi-analytic control. The paired service P is slower than inherited L in all 48 matched clocks. Every one of the 32 declared ReLU/full-conventional-menu price intersections across all three clocks is empty, including the intersections of upper not-ruled-out conditions. Thus the achieved contribution is an action-relevant Bellman comparison and original-policy correction; this equal-feature experiment supplies no neural-specific policy or adoption advantage.

The completed evidence for this item is in Table AE.I, Table AE.II, Table AH.VIII; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** The executed study now compares VALUE and MARGINAL within identical paid information, features, coefficient bounds and qualification services, and compares those actors with the full conventional menu. All 512 primary ReLU–hinge raw-proposal pairs and all 512 delivered-action pairs are equal; the hinge pays the ReLU acquisition and required audit costs. All 32 primary ReLU/full-menu sufficient and upper not-ruled-out price regions are empty, as are all 160 individual timing regions and all 32 five-clock intersections. Changing the learning objective therefore supplies no neural-specific policy or adoption advantage here. The marginal regularity, finite fitting and action-path results apply to every eligible representation and are stated as shared mathematics.

**Retained R121 remaining scientific requirement.** A neural-specific contribution still requires a strict economic or information benefit against methods with the same features, information, feasible policies and full resource bill. The present empty recorded-menu regions do not prove impossibility for every economy, network, or deployment criterion.

**Retained R121 execution and source route.** Table X.I, Table AA.VII, and the exact equal-feature rows. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** We keep the original Neural Bellman Operator, continuation critic and feasible actor as the objects being constructed and compared. The new contribution is a decision calculation that starts from an original Bellman certificate already paid for by each representation. Theorem 6 identifies the exact derivative information in its quadratic support and value enclosure; Theorem 7 converts that information into a feasible original-policy replacement satisfying the same regret and objective-width tolerances. The source implements the identical six-candidate rule for quadratic, ReLU, an exactly matched ordinary hinge, spline, degree-two and degree-three polynomial, grid and online methods. Historical model fitting, features, paid labels and full acquisition allocations stay bound to the same 56 actual actors. The neural label supplies neither an extra original-risk query nor a mathematical exemption. Equal-feature ReLU/hinge decisions and complete conventional price menus are

explicit controls. Theorem S.24 now gives the exact finite-horizon same-incumbent gain recursion for the original conditional risk operator. A nonnegative advantage certified at every admissible state propagates through a centered lower operator with nonnegative dual slack. Proposition S.25 gives its finite complete-menu minimum-occupation calculation, and Proposition S.26 states the discounted bounded-measurable fixed-point version. The proof separates physical from risk-distorted occupation and requires a valid lower bound on the entire dual infimum. It changes no current experiment and proves neither neural exclusivity nor a new general policy-iteration convergence rate.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** This establishes a shared action-level mechanism within NBO. A neural-specific economic or information advantage still requires strict improvement over methods with the same information, features, feasible policies and full resource budget. Own-incumbent gains, inexpensive internal fitting and a nonempty conditional price scenario do not establish that proposition.

### 0.2 B2. The information-complexity result is too narrow for the platform claim

**Original advisory request.**

> The theorem is a specialized deterministic one-dimensional oracle result. It does not establish randomized, multivariate, nonuniform-law, or end-to-end Bellman complexity lower bounds.

**Completed R120 author response.** The new constrained-AVaR theorem supplies the original continuation gradient modulus and the action-distance modulus $E$. The fixed-known-tail information result has matching width order

$$\min\{KR(1-\alpha)N^{-2},\ ER(1-\alpha)N^{-1}\}.$$

One exact scalar value is one query; the claim is deterministic and univariate. The original smooth model also gives the stronger separate-width coefficient $2M_\alpha R^2(1-\alpha)^2$. The nonsmooth lower-bound adversary therefore does not establish an original-primitives or smooth-class lower bound. Full policy comparisons pay both risk deficits, outward point uncertainty and the unchanged future allowance. In execution the mixed upper is necessary at two recorded accepted levels for one common Poly3 candidate at $m = 4$; no wall-time gain or optimal general Bellman information complexity is inferred from that event. The full positive-margin refinement and future-information-floor propositions state when the interface can terminate successfully.

The completed evidence for this item is in Table AH.III; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** The new constructive upper result obtains outward action-marginal labels from integrated original-risk value enclosures, with explicit finite-difference, risk-level, capacity and rounding allowances. The two actually acquired budgets contain 650 and 2,106 original root-$Q$ contexts across the two risks, respectively, rather than

free derivative labels; 1,413,828 terminal label contexts are charged in total. VALUE receives the same complete bundle as MARGINAL. These executed information counts make the interface cost visible. They do not enlarge the class covered by the inherited deterministic univariate fixed-tail lower bound. The original-formula online method remains eligible and its actual cost is retained.

**Retained R121 remaining scientific requirement.** No randomized, multivariate, nonuniform-law, formula-oracle, bit-complexity or end-to-end Bellman minimax lower bound is inferred from the earlier scalar oracle construction. Matching lower and upper bounds for the smooth original economy remain open at a separately specified information interface.

**Retained R121 execution and source route.** Proposition S.5, Table AA.III, and the acquisition rows. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The retained continuous-risk theorem is kept with its deterministic, point-value, univariate, fixed-alpha information interface. R122 adds a different, explicit information problem: what can an optimizer infer about the derivative from a paid quadratic lower support and an upper value bound at a constrained action? The sharp endpoints are attained by constant-curvature witnesses for an exact supplied slope, with the missing boundary direction unbounded absent additional information. The robust upper action model is sharp only under the stated interior, exact-information conditions. The implementation uses a finite rational probe grid and intersects separately proved economic derivative bounds, so it is not assigned sharpness it has not earned. The comparison now distinguishes a global risk-integration query problem, a local paid-support decision problem, and an empirical service bill. Proposition S.22, derived after F with its own source identity, now supplies a matching decision guarantee at the exact interior paid-support interface. For any announced randomized displacement, one admissible constant-curvature objective attains the expected robust contrast, including the nonnegative variance cost. With an otherwise arbitrary bounded future-policy discrepancy, the exact additional term is $\delta \Pr\{Z \neq 0\}$; a deterministic robust replacement or retention attains the matching implemented-policy guarantee $\max\{0, \Gamma - \delta\}$. Neural, hinge, spline, polynomial and unfitted methods have the same guarantee given this same information and feasible actions. This is a conditional information bound with a complete proof, not an assertion that its adverse functions are generated by the investment economy or that the finite qualified service is minimax. Theorem S.24 now gives the exact finite-horizon same-incumbent gain recursion for the original conditional risk operator. A nonnegative advantage certified at every admissible state propagates through a centered lower operator with nonnegative dual slack. Proposition S.25 gives its finite complete-menu minimum-occupation calculation, and Proposition S.26 states the discounted bounded-measurable fixed-point version. The proof separates physical from risk-distorted occupation and requires a valid lower bound on the entire dual infimum. It changes no current experiment and proves neither neural exclusivity nor a new general policy-iteration convergence rate.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** The retained error-versus-information rate does not cover arbitrary adaptive derivative oracles, random algorithms, high-dimensional innovations or all tail levels uniformly. The local information result does not prove a global computational lower bound for the complete NBO solver. A unified broader interface remains a substantive research task.

### 0.3 B3. The representation result is not decision-theoretic

**Original advisory request.**

> The ridge example separates restricted approximation classes but does not separate policies, service work, or economic outcomes. A conventional method with the same hinge is equivalent.

**Completed R120 author response.** The new comparison authorizes changes of the original investment policy. It treats a difference of convex continuations as Lipschitz, without assuming that the difference is convex. The independent original-policy and all-action-optimum witnesses now provide value, gain, regret and action-distance bounds for every returned decision. For ReLU at $\alpha = 1/2$, P's mean original-policy gain over its own completed incumbent is approximately $[212.675, 213.885] \times 10^{-6}$; the mean-regret upper bound falls from $215.906 \times 10^{-6}$ to $2.626 \times 10^{-6}$. The RMS action-error upper bound falls from 0.01470 to 0.001621. L returns the same ReLU policies, and conventional hinge is identical as well. These are original-investment improvements with a shared correction; they neither establish a neural geometry advantage nor borrow economic improvements from the separate posted-price example.

The completed evidence for this item is in Table AE.I; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** The new theorem controls the integral of continuation-marginal error along feasible action paths, treats feature derivative jumps explicitly, and keeps the state anchor when composing recursive values. The stationarity proposition converts an outward original-objective KKT interval into regret and action-distance bounds. The numerical distinction is now decisive: the finite-node marginal residual falls in all 28 paired treatments, but the independently enclosed 64-state mean gain $J_{\mathrm{VALUE}} - J_{\mathrm{MARGINAL}}$ is strictly negative in every treatment; 889 of 1,792 final action pairs differ. Thus fitting the reported node criterion better does not establish better decisions. The independent tables report original objectives, regret and action-error upper bounds, not only representation error.

**Retained R121 remaining scientific requirement.** The off-design feature-range cover is a proved conditional construction but was not numerically certified for the fitted actors. Neither the node residual nor the finite optimizer gap can be substituted for that missing global premise, and no neural geometry separation follows from the shared hinge representation.

**Retained R121 execution and source route.** Theorem S.7, Proposition S.8, Table X.I, and Table AA.I. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The representation-to-decision path now uses the original objective at the actor's actual proposed action. Section 5.3 retains the actor's anchored continuation, outward marginal labels and finite-box fitting results, including the unexecuted global feature-cover premise. Section 6 purchases no new feature cover: it uses the original certificate's support gap, actual support-slope interval, original curvature and capacity to enclose the original derivative. It recomputes gain, regret and objective width at every realized rounded candidate. Thus the economic claim follows from an original-action inequality and the unchanged future controller, rather than from a function approximation norm or the network architecture. Theorem S.24 now gives the exact finite-horizon same-incumbent gain recursion for the original conditional risk operator. A nonnegative advantage certified at every admissible state propagates through a centered lower operator with nonnegative dual slack. Proposition S.25 gives its finite complete-menu minimum-occupation calculation, and Proposition S.26 states the discounted bounded-measurable fixed-point version. The proof separates physical from risk-distorted occupation and requires a valid lower bound on the entire dual infimum. It changes no current experiment and proves neither neural exclusivity nor a new general policy-iteration convergence rate.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** A small fitting error alone still gives no original action or policy guarantee. The finite paid-support algorithm need not resolve every promising action, and its conditional precision result requires verified support-gap and slope-width rates. No global neural convergence or representation-specific decision advantage is asserted.

### 0.4 B4. The certificate-aware objective fails its intended service connection

**Original advisory request.**

> It improves residual span in 36 of 36 cells but worsens aggregate rejections and most quadratic cells. The missing theorem is the connection from the trained objective to action-relevant resolution.

**Completed R120 author response.** The acquired information now targets the original continuation difference at two feasible actions. A paid common-tail mismatch and a strict complete-policy gate connect this information to a permissible decision change, with the true future-controller allowance included once. Holding the three inherited support-gradient proposal rules fixed, P changes 391 of 1,024 actions and accepts 815 steps; L changes 388 and accepts 807. Their final actions differ in only 12 cases, and none of the 16 cell-mean comparisons resolves a strict incremental P advantage. The same paid pair primitive is supplied to the stronger direct conventional proposal. All fitted critics still use the ordinary original-$V_1$ value-label objective. We therefore complete an action-dependent certificate and its original-policy evaluation, while retaining action-aware training and acquisition as a distinct unresolved scientific requirement.

The completed evidence for this item is in Table AE.I, Table AH.III; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** We implemented and independently audited the replacement objective, rather than defending the previous internal span. MARGINAL minimizes a bounded finite-node original-action risk-marginal error; its 28 exact primal/dual gap upper bounds lie between approximately $1.13 \times 10^{-9}$ and $2.58 \times 10^{-8}$. All 28 node residuals improve, yet first rejections increase from 60 under VALUE to 294 under MARGINAL, all 28 independently enclosed sample mean gains are negative, and complete primary cohort service is slower in 24 of 28 comparisons. The fixed joint population intervals all contain zero. This is new adverse evidence about the proposed learning-to-service connection, not a positive result borrowed from R120.

**Retained R121 remaining scientific requirement.** The intended economic improvement is not established by this experiment. A useful next algorithm must control the off-design action paths and qualification burden, with its acquisition and certification costs paid; the present finite training certificate does not certify that improvement or a positive population gain.

**Retained R121 execution and source route.** Proposition S.6, Table X.I, Table AA.II, and Table AA.IV. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** R122 changes the delivered decision service. A completed incumbent is independently recomputed and paid within each of control, recycled and adaptive service. Recycled service attempts a strict original-policy improvement from information already computed at that incumbent. Adaptive service buys only the predeclared 256/512-cell support refinements, skipping levels no finer than the information already paid for. Each candidate must pass strict gain, original optimum regret and objective-width tests together; all refusals and all paid starts remain charged. The future-policy discrepancy is bounded but need not be differentiable or convex. This is an explicit economic safeguard, with a proof of finite accepted-step composition, rather than another internal training objective. Theorem S.24 now gives the exact finite-horizon same-incumbent gain recursion for the original conditional risk operator. A nonnegative advantage certified at every admissible state propagates through a centered lower operator with nonnegative dual slack. Proposition S.25 gives its finite complete-menu minimum-occupation calculation, and Proposition S.26 states the discounted bounded-measurable fixed-point version. The proof separates physical from risk-distorted occupation and requires a valid lower bound on the entire dual infimum. It changes no current experiment and proves neither neural exclusivity nor a new general policy-iteration convergence rate.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** A useful certified local replacement can coexist with unresolved independent confirmation at the evaluator's finite precision or poor complete comparative cost. The new service must be judged on those outcomes and strong conventional alternatives. The declared finite ladder is not a general optimal precision-allocation policy.

## 0.5 B5. Complete work favors a strong conventional method

**Original advisory request.**

> Quadratic least squares is faster, uses fewer proposal queries, and has no neural acquisition overhead. R97 therefore does not shift the computational frontier toward NBO-specific representation.

**Completed R120 author response.** The completed four-service/eight-arm comparison retains all 4,096 primary and 6,144 timing decisions and charges actual common-label acquisition, each standalone fit and audit, full hinge acquisition, loading, proposals, tail witnesses, precision ladders, rejected work, fallback and cohort bookkeeping. Summing complete standalone 64-state costs over the same 16 cells gives 94.478 seconds for L, 132.729 for P and 80.302 for direct refinement D. P purchases fewer compiled terminal rows than L, yet is slower in every primary cell and all 48 matched fresh clocks. D purchases substantially more terminal rows and is faster than P in all those clocks. The independent rational study costs a separate 1,806.153 seconds and remains charged as research validation. These complete scopes replace a point-count-only competitiveness claim; they retain the stronger traditional method and every adverse clock.

The completed evidence for this item is in Table AE.II; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** The complete comparison includes quadratic, degree-two and degree-three polynomial, spline, grid, equal-feature hinge and direct original-formula online controls. Each learned pair pays for the same acquired label bundle and its own fit and required rational audit; the hinge also pays its parent ReLU charge. All rejected proposals, conventional fallback, serialization and complete service clocks remain charged. The exact frontiers contain 1,392 volume-cost rows and 51,072 pairwise price constraints, including four prespecified reuse volumes. No ReLU full-menu sufficient region or upper not-ruled-out region survives in any primary or individual timing scenario. The conventional alternatives therefore remain essential to assessing whether learning buys an economic service.

**Retained R121 remaining scientific requirement.** A finite recorded-menu cost frontier is not a universal computational lower bound. Acquisition allocations are derived from measured process components, not separate observed standalone fits, and the independent all-policy research validation is also reported in the study ledger even when it is not a deployment charge.

**Retained R121 execution and source route.** Table AA.IV, Table AA.V, Table AA.VII, and the complete research ledger. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** Every conventional family receives the same control, recycled and adaptive services and the same original economic tolerances. The frozen full standalone bill includes historical labels, fitting and training audit for the actual inherited actor; current cold loading, original qualification, fallback, candidate arithmetic and refusal; all paid refinement attempts; serialization and process residual. The reporter separately retains incremental reuse and the full acquisition allocation. All 28,800 prescribed pairwise price constraints and

384 full-menu scenario regions remain visible, together with the 96 common-price three-clock intersections. A separate whole-study ledger records independent validation, failed attempts, reporting, QA and publication instead of allocating that research total to a favored representation.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** Recorded clocks are observations on a monitored shared host, and deployment-volume formulas are declared scenarios, not measured throughput. The interrupted first primary parent's full outer elapsed time is unavailable; its recorded nonoverlapping work is a lower bound. Neither missing costs nor favorable individual clocks may establish a complete-cost neural advantage.

### 0.6 B6. The purchase exercise is conditional sensitivity, not economic identification

**Original advisory request.**

> No learned service is adopted over quadratic least squares. Prices and fees are external, clocks are descriptive, and the policy interval is broad. The exercise does not change an economic conclusion.

**Completed R120 author response.** The fixed 48-group, 192-event population calculation has actually completed on the prospectively drawn 64 states. Under its stated finite-uniform population and independent-entropy assumption, the simultaneous 95% family gives positive event-staircase mean-gain lower bounds in all 42 learned own-incumbent groups, ranging from $2.661 \times 10^{-6}$ to $14.192 \times 10^{-6}$. The six online groups have zero observed gains and zero lower bounds; the ordinary bounded-mean lower bounds alone are zero. The reported price sets use actual paid work and the declared conventional menu. Positive own-incumbent price regions coexist with empty ReLU/full-menu regions. No calibrated monetary value, preference parameter, market demand or population adoption conclusion is supplied by those normalized-objective calculations.

The completed evidence for this item is in Table AE.III, Table AH.VIII; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** The completed economic comparison stays with the same original constrained investment objective and continuous innovation law. Reported prices value a second of computation in normalized objective units and solve the stated strict gain-minus-price-times-extra-cost inequalities exactly. The four reuse volumes and the full conventional menu were fixed before outcomes. The resulting 32 primary, 160 individual-clock and 32 all-five-clock ReLU comparisons have empty sufficient and upper not-ruled-out regions. Their meaning is a conditional decision-service comparison on the specified state cohorts, with exact endpoint and strictness flags, rather than an externally identified willingness to pay.

**Retained R121 remaining scientific requirement.** The exercise does not identify welfare prices, an empirical adoption rate or a calibrated firm-level technology choice. Upper not-ruled-out intersections are only pairwise feasibility conditions, not jointly attainable adoption claims. A consequential calibrated application still requires its own data, information model and validation.

**Retained R121 execution and source route.** Table AA.VII and the exact full-menu price constraints and menus. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The economic comparison retains the original constrained investment cost and the symbolic price criterion $NJ + pC(N)$. The price domains state strict endpoints and compare against the full conventional menu. Sufficient domains use lower economic gains; upper not-ruled-out domains express only lack of exclusion, not joint attainability. Their primary economics use all 32 states and each timing scenario uses its matching first 16. These were the written pre-sample rules. The published reporting repair restores the first-16 rule and preserves all uncorrected frozen-code tables as diagnostics. The unchanged primary-32 tables are checked exactly. The new priced-stopping proposition additionally states a narrow single-decision dominance region when every permitted continuation must first pay a compulsory cost.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** No willingness-to-pay parameter or calibrated field adoption decision is identified. Reuse, common information and uncertainty about actual computation prices can change an economic stopping problem. The compulsory-cost proposition does not solve that general problem, and the reported menus remain conditional sample-and-hardware comparisons.

### 0.7 B7. The new 84 actors lack direct population policy-cost inference

**Original advisory request.**

> R97 transfers only the uniform deterministic qualification bound to the new services. The R96 statistical intervals belong to different frozen actors and cannot establish the new actors' mean-cost ordering.

**Completed R120 author response.** The source F, actual actors M and single 640-byte state draw D are separately and actually committed before evaluation. The recorded 64 exact 40-bit dyadic states are shared across all arms and risks, and singleton initial-state services prevent a batch fallback from making one row's rule depend on another row. All 4,096 original-policy comparisons and all 48-group/192-event statistics completed under these commitments. The 95% guarantee uses the fixed one-sided binomial allocation and bounded-mean allocation from the frozen protocol; it conditions on the declared independent uniform entropy law and fixed actors. This supplies the new R120 population evidence. It neither completes an old R97 actor family nor imports R98/R115 actors, reserved states or inference into this experiment.

The completed evidence for this item is in Table AE.III; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** The current 56 stored actors and both independent fit audits were published at M before the new 640-byte entropy draw; D then fixed 64 exact two-dimensional states before policy outcomes. Independent original-primitive evaluation covers all 1,301 distinct final-policy and raw-proposal keys and all 3,712 policy comparisons. The prespecified simultaneous family contains two optimum means, 58 regret means, 28 signed learning gains, 112 signed threshold events and 116 one-sided rejection endpoints, with joint coverage at least 95% under its stated sampling assumptions. All 28 signed population mean-gain intervals contain zero, despite all 28 enclosed sample means being negative. No R97 actor or R120 sample result is transferred to these new actors.

**Retained R121 remaining scientific requirement.** The 64 independent states are the statistical units; risks, methods, matched hinges and five repeated clocks do not multiply sample size. Population coverage is conditional on the frozen actors and the stipulated ideal entropy model for the 40-bit product-uniform state law. Cross-representation population adoption and population superiority are not established.

**Retained R121 execution and source route.** Table X.I, Table AA.I, and the complete fixed population-inference rows. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The new experiment samples 32 independent dyadic initial states only after F and publishes D before any policy evaluation. It evaluates the entire fixed 58 risk/actor catalogue under all three services. Independent original-policy enclosures support 116 within-actor gain means, 232 predeclared threshold-event bounds and 348 signed conventional-comparator mean comparisons. Section 18 gives the joint 95% argument, the exact state population, the uniform policy-qualification premise, and the distinction between numerical interval observations and theorem-supported intersections. It also explains why exact action identity is a sample paired identity unless the complete method key is identical. Historical R97/R120/R121 actors or observations do not substitute for this sample. Theorem S.24 now gives the exact finite-horizon same-incumbent gain recursion for the original conditional risk operator. A nonnegative advantage certified at every admissible state propagates through a centered lower operator with nonnegative dual slack. Proposition S.25 gives its finite complete-menu minimum-occupation calculation, and Proposition S.26 states the discounted bounded-measurable fixed-point version. The proof separates physical from risk-distorted occupation and requires a valid lower bound on the entire dual infimum. It changes no current experiment and proves neither neural exclusivity nor a new general policy-iteration convergence rate.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** Population inference is for the specified dyadic initial-state law and predeclared fixed methods. Continuous economic innovations are unchanged. A transfer to a continuously uniform initial-state population requires a separate implemented-policy regularity or acquisition argument. Timing repeats, repeated audits and the old interrupted clock add no population observations.

## 0.8 B8. Independent verification stops before fitting

**Original advisory request.**

> Feature selection, interval-aware fitting, rounding, domain validation, and serialization are not independently reproduced. This is the critical new acquisition stack.

**Completed R120 author response.** The separately implemented rational validator actually evaluates all 861 unique returned policy keys, with 111,069 terminal contexts and 4,096 complete-policy comparisons. Its true Lagrange affine supports also give an original all-action optimum lower witness, rather than attaching an unproved slope to a scalar lower value. Thus the independent calculation covers regret and action-distance bounds as well as policy value. All 4,096 intervals overlap: 726 production intervals contain the independent enclosure, while 3,370 narrower production intervals are contained in it. Only the former direction inherits coverage from the independent enclosure; overlap or the latter direction does not independently prove fine production endpoints. The validator does not import the production terminal compiler, retains its exact rational traces and 1,806.153-second cost, and claims neither external authorship nor a complete historical fitting-optimizer replication.

The exact completed source/evidence route includes `results120/independent-attempt01/EXECUTION120.json`; the current response map binds all corresponding files.

**Completed R121 mathematical, algorithmic and empirical action.** Independent verification now begins at the actual finite fitting problem. A separate standard-library rational implementation reconstructs the stored feature values and derivatives, coefficient rounding, objective intervals and primal/dual witnesses for all 56 actors, executing 771,005 assertions. There are 48 actual optimizer fits and eight exactly matched hinge copies, which are not counted as independent fits. For the 28 MARGINAL actors, the finite-box minimax gap is enclosed by exact rational bounds whose largest upper endpoint is approximately $2.58 \times 10^{-8}$. The VALUE least-squares audit is reported separately. After fitting, an independent original-primitive evaluator performs 1,406,549 assertions over 167,829 terminal contexts and all delivered and first-proposed policies.

**Retained R121 remaining scientific requirement.** The audited problem has fixed features and a prespecified coefficient box. It does not certify globally optimal hidden features, the best unrestricted approximant, the unexecuted global feature cover, or the rounded future controller by exact policy equality. Independence here means a separate implementation, not external human authorship.

**Retained R121 execution and source route.** Proposition S.6, Table AA.III, and the independent fit and policy execution receipts. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** R122 performs no new fitting: all 56 model files are checked against actual M121 and the entire frozen dependency closure. The prior exact finite-box fitting and independently implemented training audits, including their limitations, remain in the supplement and inherited evidence. The new independent stage first checks every serialized support/candidate implication and then uses a separate exact-rational implementation

of the original primitives for all unique final, baseline and raw actor actions. It applies the fixed 64/128/256/512-cell accepted-claim ladder without selected exceptions. The original independent attempt's training-label/numeric-loss schema failure is retained; the separately published adapter changes only a temporary shallow numeric alias after strict equalities, preserving the training label and unchanged mathematical audit counters.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** Original primitive checks, finite fitting audits, successful package replay and independent authorship are distinct. R122 does not add a fresh independent refit, and the author's rational evaluator is not an external referee. Accepted claims unresolved after the fixed ladder remain unresolved rather than being called independently confirmed.

### 0.9 B9. The active submission remains cumulative and disproportionate

**Original advisory request.**

> A 47-page article, 37-page supplement, and 230-page appendix are excessive for the narrow current theorem and negative acquisition experiment. Historical preservation need not determine the active submission's length.

**Completed R120 author response.** The active main argument now follows the original continuation/actor problem, the action-dependent modulus, a usable complete-policy improvement rule, the executed matched comparison and its economic interpretation. The supplement contains every new proof and the independently implemented original-optimum construction. The development tables retain all 64 objective/cost cells, 192 clocks, 192 threshold events, 992 sample contrasts and every prescribed price comparison. The 35 original report bodies and the 1,049 inherited item/parent routes remain accessible with exact R119 source bindings. Their presence does not replace the new evidence: the main text reports P's adverse timing, unresolved incremental economic advantage and the stronger conventional proposal alongside its valid gains and the one observed mixed-bound acceptance.

The completed evidence for this item is in Table AE.I; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** The active account is organized around the original investment decision, integrated-risk marginal learning, its complete mathematical assumptions and the executed economic comparison. The supplement carries complete new proofs, including derivative jumps, boundaries, action-path transport, stationarity and recursive anchors; development material carries the full 58-policy, 28-contrast, clock, cost and price tables. Historical theorems, corrigenda, raw evidence and all 35 original reports remain source-bound. The exact 30-item current response retains every original quotation and every complete R120 map record. The 1,049 inherited item/parent routes remain recoverable without being counted as 1,049 independent or closed scientific issues.

**Retained R121 remaining scientific requirement.** Source organization is not evidence that the unresolved neural or economic contribution has been achieved. Final document sizes and source/PDF identities are read from the actual build and publication records. The dated theorem/code registry is a structural index, not a new proof-by-proof revalidation of every historical source.

**Retained R121 execution and source route.** The current source structure, `CONTENT_RETENTIO N_MAP121.md`, `SOURCE_READY121.json`, and `EVIDENCE_MANIFEST121.json`; later source-specific CI and publication receipts have their own identities. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The main article is organized around the original economic problem, the Bellman/continuation/actor construction, the new support-to-action information theorem and safeguard, and the complete original-policy experiment. Complete proofs stay in the technical supplement. Historical main-article accounts, numerical programs, diagnostics, failed runs, negative results and older response routes remain in the development companion with their original source identities. The retention map separates current main contributions, technical dependencies, inherited evidence and independently constrained episodes. The controlling report and all 30 exact current requests are linked to the full preceding responses instead of silently discarding their scientific content. Theorem S.24 now gives the exact finite-horizon same-incumbent gain recursion for the original conditional risk operator. A nonnegative advantage certified at every admissible state propagates through a centered lower operator with nonnegative dual slack. Proposition S.25 gives its finite complete-menu minimum-occupation calculation, and Proposition S.26 states the discounted bounded-measurable fixed-point version. The proof separates physical from risk-distorted occupation and requires a valid lower bound on the entire dual infimum. It changes no current experiment and proves neither neural exclusivity nor a new general policy-iteration convergence rate.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** Moving material changes presentation, not proof or empirical status. A clean source/PDF build and a compact main narrative do not themselves resolve open scientific novelty, scaling or application questions. The historical companion remains deliberately extensive because deletion would destroy the requested research record.

### 0.10 B10. The negative result is more compelling than the platform claim

**Original advisory request.**

> The strongest new fact is that a valid global residual objective can fail to improve delivered decisions. The manuscript should either make that misalignment central or supply a new action-aligned objective and theorem.

**Completed R120 author response.** The new method buys information about an original action contrast and can change the decision. The continuation modulus, fixed-tail information bounds and quantitative tail-defect/refinement analysis are proved; the completed-incumbent gate and the three support proposals retain their R108–R110 ownership. The completed diagnostics distinguish use from independent contribution: the mixed upper is active at 5,961 of 9,016 paid levels and necessary at two accepted levels representing one Poly3 candidate shared by P and D. The transferred incumbent tail-defect bound is inactive in all 2,290 acquired density pairs. P's final action differs from L in 12 cases, with nine exact cell identities and seven unresolved mean differences; its full cost is higher in all 16 primary cells. We claim the proved comparison and actual fixed-resolution decision example, while preserving the unresolved incremental competitiveness and the unfinished action-aware acquisition problem.

The completed evidence for this item is in Table AH.III, Table AE.II; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** The new experiment produces a sharper scientific obstruction inside the NBO problem: accurately solving an action-marginal finite training objective still fails to improve the delivered original decision. All 28 node residuals fall, all 28 sample mean gains are negative, first rejections rise to 294 from 60, and the full primary cohort service is slower in 24 comparisons. At the same time, all 3,712 delivered policies satisfy the prespecified production qualification and independent evaluation records their original regret and action-error bounds. This separates successful finite fitting and a functioning safeguard from the unachieved learning advantage. The action-path and KKT results specify the missing off-design quantities that a stronger method must actually control.

**Retained R121 remaining scientific requirement.** These adverse findings do not complete the original ambition of a computationally competitive neural Bellman operator. Global action-path certification, cost-effective precision allocation and a strict equal-information economic advantage remain research obligations. The paper retains its original problem and Econometrica objective while documenting this unmet mechanism.

**Retained R121 execution and source route.** Table X.I, Table AA.II, Table AA.IV, and Theorem S.7. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** We use the earlier negative economic results to identify a new actionable defect: an actor can fit a marginal objective more accurately without producing a useful original-policy improvement, while an already paid original certificate can contain unused decision information. The new theorem extracts exactly that information, and the source tests whether its safe use affects the original policies and full cost. Earlier conventional wins, empty adoption regions and ReLU/hinge equivalence remain visible. The manuscript continues to develop Neural Bellman Operators for the specified recursive decision problem and retains the Econometrica numerical-methods objective. Theorem S.24 now gives the exact finite-horizon same-incumbent gain recursion for the original conditional risk operator. A nonnegative advantage certified at every admissible state propagates through a centered lower operator with nonnegative dual slack. Proposition S.25 gives its finite complete-menu minimum-occupation calculation, and Proposition S.26 states the discounted bounded-measurable fixed-point version.

The proof separates physical from risk-distorted occupation and requires a valid lower bound on the entire dual infimum. It changes no current experiment and proves neither neural exclusivity nor a new general policy-iteration convergence rate.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** The local algorithm is a contribution to the research program, not a declaration that the platform's neural-specific advantage has been established. Both positive own-incumbent outcomes and negative comparative outcomes must determine the next mathematical and computational questions.

### 0.11 M1. Recast the paper around one method-neutral question

**Original advisory request.**

> Ask when a stored continuation can be safely purchased for recursive-risk dynamic decisions. Treat neural networks as one optional representation, not the source of validity.

**Completed R120 author response.** Neural Bellman Operators remains the same original-economic-decision author project. The main question is how learned continuations, recursive feasible actors and action-relevant information can produce valid improvements at justified complete cost. The investment model now supplies actual policy changes, original-value/regret bounds and fixed-law gain inference, rather than an internal fit metric alone. Its continuous shock changes stock imbalance, while investment trades current adjustment cost against the original risk-sensitive continuation and capacity constraint. The main interpretation also confronts the strong online and direct controls and full paid work. The new safeguard is shared across representations; those facts preserve the neural Bellman research identity without assigning an unsupported architecture-exclusive mechanism.

The completed evidence for this item is in Table AE.I; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** The author retains Neural Bellman Operators and the Econometrica numerical-methods objective. The scientific concentration requested here is implemented within that research identity: can learning the marginal of the integrated original continuation risk improve the feasible original actor after qualification and full information costs? The revised theory and the frozen VALUE/MARGINAL experiment directly address that question. The executed answer is adverse for this design: every paired sample mean favors VALUE, all fixed population intervals remain unresolved, and the ReLU full-menu price sets are empty. Both the shared methodological contribution and the unachieved neural improvement are stated explicitly.

**Retained R121 remaining scientific requirement.** Literal redirection to a different paper or venue is not followed. A negative result is not relabeled as completion of the neural economic objective; the remaining action-path, information-efficiency and adoption questions continue inside the original model.

**Retained R121 execution and source route.** The main theoretical sequence and Table X.I; the unchanged original report and complete prior treatment remain above. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The author revision focuses the main argument on a concrete question within the original NBO program: when can original Bellman information already purchased for an acquired actor produce a certified, economically useful action change? Continuation critics, feasible actors, recursive objectives and strong conventional comparators remain central. The title remains Neural Bellman Operators, and the paper retains its Econometrica numerical-methods goal. The response adopts the request for a precise, testable economic contribution while preserving the authorized research identity and the previous models and results. Theorem S.24 now gives the exact finite-horizon same-incumbent gain recursion for the original conditional risk operator. A nonnegative advantage certified at every admissible state propagates through a centered lower operator with nonnegative dual slack. Proposition S.25 gives its finite complete-menu minimum-occupation calculation, and Proposition S.26 states the discounted bounded-measurable fixed-point version. The proof separates physical from risk-distorted occupation and requires a valid lower bound on the entire dual infimum. It changes no current experiment and proves neither neural exclusivity nor a new general policy-iteration convergence rate.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** A method-neutral safeguard does not erase the need to establish what the neural representation contributes. That question is addressed by equal-information controls and economic comparisons, not by changing the original objective, information set, venue or topic.

### 0.12 M2. Establish the true novelty of the information theorem

**Original advisory request.**

> Give a theorem-by-theorem comparison with information-based complexity. Add a genuinely new randomized, nonuniform-law, multivariate, or Bellman-compositional result, or present the current theorem as a specialized supporting lemma.

**Completed R120 author response.** The novelty statement distinguishes the new original constrained-AVaR smoothness calculation, action-distance integral interface, matching fixed-tail query order and quantitative constrained-tail transfer from the inherited common-risk identity and strict policy gate. The original model's additional smoothness gives its own stronger upper coefficient and is excluded from the generic nonsmooth lower-class claim. The full finite-resolution theorem includes paid deficits and the true future-policy floor. Actual execution shows one Poly3 acceptance at a paid resolution that the separate convex upper cannot authorize at that same resolution. It does not measure a traditional method after another refinement, a counterfactual

time saving, or general end-to-end Bellman optimality. The economic and timing comparisons retain those precise limits.

The completed evidence for this item is in Table AH.III; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** The revised novelty statement separates known ingredients from the original-risk construction. Action-null quotients, bounded convex readouts, dual witnesses and KKT regret bounds retain their historical and classical attribution. The new proof derives constrained integrated-AVaR marginal regularity from the investment primitives, obtains paid outward probe labels, verifies the finite stored problem, and carries interval marginal errors through nonsmooth feature ranges into feasible action comparisons while preserving recursive anchors. The actual 2,756 original root-$Q$ contexts and separate fit audits make that construction executable. The experiment establishes no optimality of its information complexity and no neural-exclusive representation gain.

**Retained R121 remaining scientific requirement.** The inherited deterministic point-value univariate fixed-$\alpha$ lower bound does not become a general Bellman lower bound through this application. Matching bounds at the original-formula interface and a decision-sensitive learning guarantee beyond the stated cover assumptions remain unproved.

**Retained R121 execution and source route.** Theorem S.4, Proposition S.5, Proposition S.6, and Theorem S.7. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The new information theorem states an explicit finite payload and supplies extremal witnesses for its derivative endpoints, including one-sided capacity boundaries. The proofs identify when the robust action model is sharp and when additional economic bounds or outward finite probes make the implemented model wider. The main text positions the support calculation relative to established inexact-oracle and smooth-convex interpolation work, with the original Bellman payload, implemented-future allowance and qualification interface as its specific application. The earlier deterministic point-value univariate fixed-alpha continuous-risk lower bound remains fully retained with its original proof and scope. Proposition S.22, derived after F with its own source identity, now supplies a matching decision guarantee at the exact interior paid-support interface. For any announced randomized displacement, one admissible constant-curvature objective attains the expected robust contrast, including the nonnegative variance cost. With an otherwise arbitrary bounded future-policy discrepancy, the exact additional term is $\delta \Pr\{Z \neq 0\}$; a deterministic robust replacement or retention attains the matching implemented-policy guarantee $\max\{0, \Gamma - \delta\}$. Neural, hinge, spline, polynomial and unfitted methods have the same guarantee given this same information and feasible actions. This is a conditional information bound with a complete proof, not an assertion that its adverse functions are generated by the investment economy or that the finite qualified service is minimax. Theorem S.24 now gives the exact finite-horizon same-incumbent gain recursion for the original conditional risk operator. A nonnegative advantage certified at every admissible state propagates through a centered lower operator with nonnegative dual slack. Proposition S.25 gives its finite complete-menu minimum-occupation calculation, and Proposition S.26 states the

discounted bounded-measurable fixed-point version. The proof separates physical from risk-distorted occupation and requires a valid lower bound on the entire dual infimum. It changes no current experiment and proves neither neural exclusivity nor a new general policy-iteration convergence rate.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** Sharpness in this declared local information class is not a novel universal complexity theorem. Broader derivative-information, randomized and multidimensional lower bounds require new adversarial classes and cannot be inferred from the current witnesses.

### 0.13 M3. Develop an action-aligned acquisition objective

**Original advisory request.**

> Construct a loss that controls rejection, endpoint work, or policy-cost resolution on a predeclared context law. Prove the connection and validate it prospectively against complete work.

**Completed R120 author response.** All 16 stored model/risk artifacts were freshly acquired under the frozen ordinary original-$V_1$ interval-midpoint value-label rule, with 256 common sites per risk and complete fitting/audit costs. The new action-relevant acquisition occurs inside the paid policy-comparison interface; it does not change that training objective. Its original-policy consequences are independently evaluated, including 42 positive learned-incumbent gain groups and the unresolved P-versus-L incremental mean comparison. The item therefore separates the completed comparison-information theorem from the still-needed action-aligned loss, acquisition theorem and newly fitted-actor validation. R98's distinct seven-objective acquisition experiment remains preserved under its own unchanged actor and sample barriers.

The completed evidence for this item is in Table AE.I, Table AE.II; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** An action-aligned acquisition target is now actually implemented: the known current-action cost is kept exact, the critic fits intervals for the original integrated continuation-risk marginal in original action units, and its bounded minimax readout has an independent finite optimality audit. VALUE receives the same probes and other paid information. This change lowers all 28 node residuals, but the final actions differ in 889 pairs and the independently bounded sample mean gain is negative in all 28 treatments. The action-path theorem explains why a node fit alone cannot provide the needed off-design guarantee; the proof requires a feature-range cover that this campaign has not certified.

**Retained R121 remaining scientific requirement.** An implemented marginal target is not yet a successful policy-improvement or acquisition-allocation mechanism. A globally certified path budget, an economically optimal stopping rule and a positive full-cost learning benefit require further algorithmic development and a separate prospectively frozen experiment.

**Retained R121 execution and source route.** Proposition S.5, Theorem S.7, Table X.I, and Table AA.III. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** R122 gives an action-aligned acquisition and stopping mechanism. Its uncertainty radius is $\Psi(e, c, r) = \inf_{0<h\leq r}(e/h + ch)$, so supporting-value error, slope uncertainty and feasible action distances enter explicitly. The finite service tests six rounded feasible displacements and selects the largest certified original-policy gain among those satisfying the same regret and width tolerances. If no candidate passes, the adaptive arm follows its fixed support-refinement ladder at the unchanged incumbent. The theory states the conditional square-root conversion from support-value precision to derivative precision, the additional qualification-width obstruction, finite gain accumulation and a compulsory-cost stopping region. Theorem S.24 now gives the exact finite-horizon same-incumbent gain recursion for the original conditional risk operator. A nonnegative advantage certified at every admissible state propagates through a centered lower operator with nonnegative dual slack. Proposition S.25 gives its finite complete-menu minimum-occupation calculation, and Proposition S.26 states the discounted bounded-measurable fixed-point version. The proof separates physical from risk-distorted occupation and requires a valid lower bound on the entire dual infimum. It changes no current experiment and proves neither neural exclusivity nor a new general policy-iteration convergence rate.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** The finite-probe implementation does not automatically attain the continuum infimum or a geometric accuracy rate. Theoretical precision allocation requires the stated oracle error/work premises. General optimal stopping with reusable information and stochastic costs remains open.

## 0.14 M4. Expand the conventional geometry frontier

**Original advisory request.**

> Compare ReLU with conventional oblique hinges, adaptive triangulation, convex regression, and representation-neutral basis selection under equal information, rank, acquisition, and validation budgets.

**Completed R120 author response.** The contemporary frontier contains online semi-analytic, quadratic, ReLU, equal-feature hinge, spline, degree-two polynomial, degree-three polynomial and grid arms, each with C, L, P and D. All 992 unordered within-risk cell comparisons are retained. Every learned completed cell is strictly worse than online in sample original value, and all 512 ReLU–hinge policy pairs are identical under full equal acquisition charges. D has a strictly better independent mean than L and P in five cells, two exact online identities and nine unresolved comparisons; it is faster than P in every matched clock. Thus conventional geometry, direct original-formula access and the shared certificate all receive their actual competitors. A favorable comparison with only a neural arm's own incumbent is not used as the representation frontier.

The completed evidence for this item is in Table AE.I, Table AH.VIII; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** The executed conventional frontier includes quadratic, degree-two and degree-three polynomial, spline, grid, ordinary hinge with the identical ReLU features and coefficients, and direct online original-formula solution. Their information, feasible actor domain and qualification tolerances are held fixed within the prescribed comparisons. The hinge pays the parent learned acquisition and audit charge and matches every one of the 512 primary ReLU raw and delivered actions. Every one of the 32 primary and 160 individual-clock ReLU full-menu upper not-ruled-out price regions is empty. This treats equivalent geometry as equivalent rather than attributing an ordinary hinge construction exclusively to a network.

**Retained R121 remaining scientific requirement.** The fixed feature menus and two acquisition budgets do not exhaust conventional approximation or neural architectures. An exclusive neural claim needs a resource-matched separation beyond these identical-feature controls; the present data supply none.

**Retained R121 execution and source route.** Table AA.III, Table AA.IV, Table AA.VII, and the equal-feature control rows. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The current experiment retains all seven acquired representation families, both losses and both budgets, plus the original online solver. ReLU and ordinary hinge use exactly the same features and coefficients; the hinge retains its parent-fit and audit acquisition charges. Every representation receives the new support-to-action rule and the same feasible rounded candidates, risk distributions, future policy and accuracy requirements. Complete conventional-menu constraints include alternative losses, budgets and services at the same risk, so a neural actor cannot be declared superior by removing an eligible conventional implementation.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** The comparison is with the complete declared menu, not all possible conventional codes. Additional strong algorithms can motivate a separately frozen study, but cannot be introduced or removed retrospectively to alter this experiment's adoption domains.

### 0.15 M5. Treat the negative result as first-class evidence

**Original advisory request.**

> Report aggregate and cell-level rejection changes beside span improvements, explain where the global objective allocates approximation effort, and do not label 36/36 span reduction as delivered-service success.

**Completed R120 author response.** The main text and complete development tables retain P's 16 primary cost losses and all 48 fresh-clock losses to L, nine exact and seven unresolved P/L mean comparisons, and all six zero-gain online population groups. P retains 2,257 rejected candidates, including 785 exhausted precision ladders; D retains 633 rejections, including 194 exhausted ladders. The inactive transferred-deficit result and the distinction between 5,961 active mixed bounds and only one necessary economic candidate are reported directly. The older R97 residual-span mismatch, adverse R119 phases, conventional wins and unverified neural-adoption regions remain unchanged. No discarded clock, unresolved sign or failed attempt is removed to make the new mechanism appear uniformly favorable.

The completed evidence for this item is in Table AH.III, Table AE.II; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** Every primary raw proposal, qualification attempt, rejection, returned policy and matched timing row is retained. The 4,066 attempts include all 354 first rejections and their conventional fallback; none is removed from costs or independent evaluation. The new adverse pattern—28 improved node fits with 28 worse sample means, 294 versus 60 first rejections, and no ReLU full-menu price region—is presented alongside all favorable state-level threshold events. The 28 mean comparisons are not independent replications, since the same 64 states and matched feature controls recur. R49, R97, R119 and R120 negative evidence remains unchanged and is not used in place of this new execution.

**Retained R121 remaining scientific requirement.** No adverse cell, unfavorable clock or failed historical attempt is converted into a favorable aggregate by reselection or retiming. Current sample signs do not establish population harm: all 28 fixed population mean-gain intervals still contain zero.

**Retained R121 execution and source route.** Table X.I, Table AA.II, Table AA.V, and the complete signed threshold rows. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The response and development companion preserve the executed R120 own-incumbent gains alongside conventional timing wins and the empty common-clock full-menu frontier. They preserve R121's improved finite marginal fits, unfavorable independently enclosed MARGINAL-versus-VALUE sample gains, increased initial rejections, equal-feature ReLU/hinge decisions and empty complete menus. R122 separately retains the interrupted primary, original independent-schema failure, original reporter failure, all adaptation receipts, every declined candidate and unresolved accepted claim. These are observations with their own sources, not overwritten outcomes.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** A negative outcome does not complete the original neural research aim, and a safe local gain does not negate previous comparative losses. The next

algorithmic question must use the economic and information bottleneck actually exposed by the complete results.

### 0.16 M6. Execute a consequential application or narrow the claim

**Original advisory request.**

> Either calibrate policy margins, compute prices, fees, and workloads in an application where adoption matters, or state that the paper is about numerical certification rather than economic purchase.

**Completed R120 author response.** The economic interpretation reports the original risk-sensitive investment objective, not a different loss or welfare surrogate. It shows how the independently bounded ReLU own-incumbent gains reduce original regret and action-distance ceilings, and why a larger spline gain can reflect a worse incumbent. The online semi-analytic policy is already strictly better than every learned completed policy on the sample, while direct refinement is a material traditional competitor. The simultaneous 42 positive learned-incumbent lower bounds and all actual price scenarios are reported with their population and cost scopes. The units remain normalized original-objective units: this two-date experiment adds decision-relevant evidence but does not claim monetary, consumption-equivalent or field-adoption calibration.

The completed evidence for this item is in Table AE.I, Table AE.III; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** The actual application remains the original two-date constrained investment problem with two state coordinates, continuous innovations, nested AVaR at $\alpha = 1/2, 3/4$, and the unchanged implemented future-policy allowance. Independent evaluation now reports original objective intervals, original-optimum support bounds, policy regret, action-error upper bounds and complete computation costs for all 58 cells. All 3,712 delivered policies meet the frozen production qualification, but MARGINAL worsens every paired sample mean and yields no ReLU full-menu adoption region. These are economically interpretable investment decisions under the original objective; they are not gains from a substitute posted-price or recursive-utility model.

**Retained R121 remaining scientific requirement.** A consequential externally calibrated application and a demonstrable full-cost neural economic benefit remain unfinished. The preserved stochastic-control, nonlinear recursive-utility and NDU/FBSDE connections retain their own hypotheses; this two-date execution neither validates those different models nor changes the intended research direction.

**Retained R121 execution and source route.** The original-economy definitions, Table AA.I, Table AA.II, and Table AA.IV. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The new service is executed in the same constrained investment economy with continuous innovations and nested original AVaR. The manuscript reports original policy cost and optimum regret, action changes and curvature-based

action-error bounds, independent certificate compatibility, full work and conditional computation-price domains. The mathematical economic links to stochastic control, nonlinear recursive utility and the Brownian NDU/FBSDE setting retain their own assumptions and proofs in the supplement. No result from a different objective or information set is relabelled as an improvement in this economy. Theorem S.24 now gives the exact finite-horizon same-incumbent gain recursion for the original conditional risk operator. A nonnegative advantage certified at every admissible state propagates through a centered lower operator with nonnegative dual slack. Proposition S.25 gives its finite complete-menu minimum-occupation calculation, and Proposition S.26 states the discounted bounded-measurable fixed-point version. The proof separates physical from risk-distorted occupation and requires a valid lower bound on the entire dual infimum. It changes no current experiment and proves neither neural exclusivity nor a new general policy-iteration convergence rate.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** The stylized investment test is not a calibrated consequential field application. Such an application remains an explicit research obligation, with identification, economic calibration, constraints and meaningful computation prices to be supplied; the paper's broader research aim is retained.

### 0.17 M7. Add a genuine scaling experiment

**Original advisory request.**

> Vary horizon, controls, innovation dimension, and risk level under matched comparators. Report product-point counts, memory, endpoint work, acquisition, and complete service work.

**Completed R120 author response.** The completed experiment is explicitly two-dimensional, two-date and evaluated at the two prescribed risk levels. It reports each service's full acquisition, original-value queries, candidate/refinement counters, raw process costs and complete resource-vector frontier. The four predeclared amortization volumes are conditions computed from these retained measurements; they are not measured throughput, higher-dimensional scaling or a horizon experiment. The new information theorem has a scalar-tail scope, and the generic lower bound excludes richer original-formula access. Existing higher-dimensional and application evidence retains its own source and limitations. Broader scaling and consequential application claims remain concrete research obligations rather than being filled by these 64 new initial states.

The completed evidence for this item is in Table AE.II, Table AH.VIII; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** Both prespecified budgets were acquired and fitted in full before the held-out states: 325 and 1,053 original root-$Q$ contexts per risk, a ratio of 81/25. The grid and spline feature counts and all actual acquisition, audit and service costs are reported at both budgets. Among the 28 coarse-to-fine original sample mean comparisons, 15 enclosed gains are strictly positive, 12 strictly negative and one contains zero.

These are descriptive comparisons on the common fresh sample, not an added population inference family. Different design-node sets also prevent a finite-node residual ratio from serving as an off-design convergence certificate.

**Retained R121 remaining scientific requirement.** The experiment is genuine information-budget scaling at fixed dimension and horizon. It does not establish dimension scaling, horizon scaling, an asymptotic rate or monotone improvement with more data; those remain separate scientific requirements.

**Retained R121 execution and source route.** Table AA.VI, Table AA.III, and both actual acquisition-process receipts. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The new study measures the complete fixed-dimensional service under two risk levels, two inherited acquisition budgets, three services and three full process clocks. It retains actual risk-integration and candidate/refinement counters, complete timing residuals, and fixed reuse volumes 32, 256, 4096 and 65536. The precision theorem states the necessary error-versus-information assumptions and width obstruction rather than deriving convergence from a timing plot. Historical horizon/vector-control statements and earlier scalability limitations remain in the companion. Theorem S.24 now gives the exact finite-horizon same-incumbent gain recursion for the original conditional risk operator. A nonnegative advantage certified at every admissible state propagates through a centered lower operator with nonnegative dual slack. Proposition S.25 gives its finite complete-menu minimum-occupation calculation, and Proposition S.26 states the discounted bounded-measurable fixed-point version. The proof separates physical from risk-distorted occupation and requires a valid lower bound on the entire dual infimum. It changes no current experiment and proves neither neural exclusivity nor a new general policy-iteration convergence rate.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** No new dimension or horizon scaling experiment has been executed. Reuse-volume extrapolations are not measured scaling results, and low-dimensional paid-support success does not establish high-dimensional neural competitiveness. A genuine scaling study requires a new prospective design with strong matched conventional algorithms.

## 0.18 M8. Strengthen timing evidence

**Original advisory request.**

> Use more repetitions, record or control processor state, and make operation counts primary. Predeclare the inferential target for deployment clocks if a probabilistic purchase claim is desired.

**Completed R120 author response.** All three 32-state timing processes completed, retaining 6,144 timed actions and 192 cohort clocks. Every timed action, interval and recorded query count matches its corresponding primary rule; no mismatch or clock was censored. P is slower than L in all 48 matched clocks; D is faster than P in all 48 and faster than L in 41, with seven losses and two adverse cell medians. Cohort measurements, standalone acquisition charges, loading and larger process/launcher scopes are labeled separately. These are descriptive engineering observations under the observed execution conditions. The clocks are not extra population observations and do not supply confidence bounds for future hardware, latency, energy prices or external load.

The completed evidence for this item is in Table AE.II; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** Five fresh timing processes execute the fixed first 32 states in the same catalogue, and all 9,280 saved timing actions match the primary state and action identities. Among 140 matched MARGINAL/VALUE cohort clocks, MARGINAL is faster in 40 and slower in 100; including the measured allocated acquisition and audit bill changes these counts to 38 and 102. In the 64-state primary cohort it is faster in four of 28 complete-service comparisons. All five clocks, process loads, serialization, rejected proposals and fallback are retained, and the selected study-process ledger includes the independent policy validation. Nested timing processes are not added again to the enclosing primary clock.

**Retained R121 remaining scientific requirement.** Five descriptive observations are not a runtime confidence interval or a claim about the machine-independent timing distribution. Allocated standalone deployment costs are calculated from measured components and are distinguished from directly observed whole processes. Repeats do not add independent economic states.

**Retained R121 execution and source route.** Table AA.IV, Table AA.V, and the selected research-stage cost rows. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The completed primary retry contains the entire prescribed catalogue and three fresh serial rotated first-16 timing processes. The original interrupted attempt and its complete clock zero and partial clock one remain preserved. The restart protocol predates the retry and requires exact identity checks for all 9,808 saved old rows on 16 economic fields. The current service allocation uses the frozen full outer-clock and cohort formula. A separate restrictive sensitivity intersects the three current clock domains with the old complete-clock domain; it cannot create an adoption region. The research ledger counts the complete primary parent once, records nested clocks as components and retains all observable failed and administrative work.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** The original interrupted parent's full time and exit cause are unknown. Its 443.899084812-second completed-cohort sum is only a lower bound; it is not

a full failed-run charge. Shared-host clocks, administrative recovery and incomplete publication timing preclude a fabricated exact whole-study total or a dedicated-hardware benchmark claim.

### 0.19 M9. Independently reimplement fitting and package a standalone environment

**Original advisory request.**

> Reproduce feature selection, interval-aware fitting, rounding, whole-domain validation, and serialization independently. Include a container or lockfile and all inherited modules needed to run the new tests from the artifact alone.

**Completed R120 author response.** The exact source F, actual actor M, new sample D and pre-outcome reporting specification R are preserved together with their real launch and publication receipts. Primary evidence was separately published as E0 before final reporting. Independent original-primitives validation, the fixed rational statistics and the first read-only reporter have all completed with exit zero; the reporter binds 777 unchanged input files and 36 stable products. The derivative algebra diagnostic preserves its first source-path failure and the corrected second attempt, without changing frozen science. The bounded package audit and source CI distinguish record/byte checks and statistical/report replays from a fresh primitive-economy rerun. Actual final source, PDFs, build logs and CI states belong to the separate source/PDF and publication declarations; no R119 success or future R120 job is asserted here as a completed R120 CI.

The exact completed source/evidence route includes `provenance/SOURCE_PUBLICATION120.json`; the current response map binds all corresponding files.

**Completed R121 mathematical, algorithmic and empirical action.** The independent fitting implementation has now executed on all 56 stored actors, checking exact feature and derivative assembly and finite primal/dual witnesses without calling the fitting solver. The separate original-primitive evaluator has completed all 1,301 distinct raw and delivered policy keys with 1,406,549 assertions. The fixed inference calculation and deterministic reporter also exited successfully. The reporter binds 473 input files, checks 220,310 record assertions and reproduces 49 stable output files without fitting, new states, original economic evaluations or timing reruns. The standalone source, dependency pins, immutable inputs and reproduction commands distinguish this bounded reporting check from a full scientific replay.

**Retained R121 remaining scientific requirement.** A separate implementation is not external authorship, and the finite fitting certificate does not optimize hidden features or establish a global feature cover. Final source-specific CI and publication evidence are distinct from the already successful scientific stages and from the M/R transport workflows; they are identified by their own receipts without a future-success assertion inside this source.

**Retained R121 execution and source route.** The independent fit/policy receipts, `SOURCE_READY121.json`, `EVIDENCE_MANIFEST121.json`, and the source-specific CI/publication records. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The dependency closure preserves actual learned models, prior fit sources and independently checked finite-box optimization records. The present source freeze binds 243 scientific members. A separately versioned independent adapter leaves the numerical primitive and audit algorithms unchanged and makes its schema-only mapping explicit. Referee-package readers check the saved original scientific receipts, raw witnesses, all statistical and price tables, failed attempts, source identities and PDF hashes. Clean reconstruction is separated from a new scientific rerun. Newly reconstructed author text after workspace loss receives a new hash; exactly recovered readers retain their verified original hash.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** No independent refit was added in R122 because no model was newly fit. A standalone package certifies reproducible custody only to its checked dependencies and commands; it cannot certify all historical claims, unseen environments or external scientific acceptance. CI success is asserted only after an actual completed run.

### 0.20 M10. Shorten the active submission substantially

**Original advisory request.**

> Move historical material to an archival companion. The active article should contain only the current qualification theorem, precise information result, acquisition objective, matched experiment, and supported economic interpretation.

**Completed R120 author response.** The main paper concentrates on the original recursive decision, complete new mathematical argument, contemporaneous experiment and economic interpretation. Full assumptions, proofs, independent optimum verification and all detailed tables remain in the supplement/development companion. All 35 exact original reports are copied unchanged into the current source package; the 620 major, 394 additional and 35 overlapping-parent treatments retain their original identities and specific remaining requirements. The current response supplies an actual proof/evidence route for each of the 30 controlling items. This organization preserves NBO's learned continuation and actor problem, the strong traditional comparisons and every valid historical result, while keeping the current contribution assessable without treating document reorganization as scientific closure.

The completed evidence for this item is in Table AE.I; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** The principal narrative now follows one economic problem, the original-risk marginal acquisition and fitting construction, its action-path and stationarity theorems, and the completed 28-treatment economic test. Full proofs reside in the supplement; all policy, fitting, budget, timing and price tables and the research history have explicit development routes. The complete original advisory quotations and R120 treatments remain in this response and its machine map. Historical proof and code sources, including the original corrigenda and negative results, remain recoverable by commit and hash rather than being silently deleted to shorten the article.

**Retained R121 remaining scientific requirement.** Final page counts are facts of the actual strict build, recorded in the retention and publication records. Structural reorganization does not close the unresolved contribution questions, and a historical syntax/index scan is not a new correctness proof.

**Retained R121 execution and source route.** The main/supplement/development wrappers, `CONTENT_RETENTION_MAP121.md`, `SOURCE_READY121.json`, and the actual source/PDF publication records. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The active main article presents the original NBO construction, the new action-information result, its original-policy safeguard and the complete economic comparison. Retained action/risk-information and prior policy-improvement statements are relocated to the supplement with complete proofs; long historical execution and response accounts are placed in the development companion. The response itself keeps exact requests and full previous treatments because it is an audit document rather than the main exposition. The content-retention map makes each move and source identity explicit. Theorem S.24 now gives the exact finite-horizon same-incumbent gain recursion for the original conditional risk operator. A nonnegative advantage certified at every admissible state propagates through a centered lower operator with nonnegative dual slack. Proposition S.25 gives its finite complete-menu minimum-occupation calculation, and Proposition S.26 states the discounted bounded-measurable fixed-point version. The proof separates physical from risk-distorted occupation and requires a valid lower bound on the entire dual infimum. It changes no current experiment and proves neither neural exclusivity nor a new general policy-iteration convergence rate.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** Editorial focus must not delete valid theorems, proofs, programs, data, failures or negative findings. Final PDF lengths and cross-references are checked after the actual result integration and strict build, not inferred from source line counts.

### 0.21 Minor 1. Error-versus-information terminology

**Original advisory request.**

> 1. State 'N^{-2}' consistently as an error-versus-information rate; "information exponent two" is easy to misread.

**Completed R120 author response.** The new result is called interval width versus scalar point information, with order $\min\{KR(1-\alpha)N^{-2}, ER(1-\alpha)N^{-1}\}$. Original-policy contrast, certified gain, regret, value width, squared action-distance bound and complete work are separate quantities throughout the proofs and all 64 objective/cost cells. The independently computed optimum is an enclosure, so the displayed RMS quantities are upper bounds rather than measured errors

to a known exact optimizer. The achieved R119 $N^{-2}$ tail-uniform AVaR width result retains its established attribution; the executed economic gains require their own policy evidence.

The completed evidence for this item is in Table AE.I; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** The inherited $N^{-2}$ statement remains an error-versus-information bound under its deterministic scalar oracle convention. The new marginal construction displays its finite probe and label tolerances separately from fitted node error, feature-cover error and action regret. The actual budgets contain 325 and 1,053 original root-$Q$ contexts per risk, and the 28 coarse-to-fine sample comparisons include both positive and negative gains. These two budget observations are reported as executed information scaling, not evidence for an asymptotic exponent or a universal neural convergence rate.

**Retained R121 remaining scientific requirement.** No rate for general Bellman information complexity, end-to-end time or economic gain is inferred from a function-class approximation rate or from the two executed budgets.

**Retained R121 execution and source route.** Proposition S.5, Theorem S.7, and Table AA.VI. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The manuscript retains $N^{-2}$ as an error-versus-information rate under the precise continuous-risk oracle. The new support calculation separately uses an error-to-derivative conversion of order $\sqrt{e}$ under its interior conditions; it does not call either statement an information exponent identifying neural superiority.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** These are different information interfaces and units. Neither rate may be transferred to the other without its stated assumptions.

### 0.22 Minor 2. Oracle qualifiers

**Original advisory request.**

> 2. Put "deterministic, point-value, univariate, fixed-alpha" adjacent to every high-level summary of the lower bound.

**Completed R120 author response.** Deterministic, exact scalar point-value, univariate, uniform-law, convex-Lipschitz and fixed-known-tail qualifiers appear with the new information statement. Purchasing the two continuation values at a point costs two queries. The original smooth investment formula is a narrower class than the lower adversary, and risk-tail localization, finite point precision, terminal solves and the future controller remain separately paid. R119's risk-level-uniform width

theorem is retained under its own assumptions. The completed two-risk experiment does not establish randomized, multivariate or arbitrary-law oracle optimality.

The completed evidence for this item is in Table AH.III; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** Every operative interpretation of the inherited lower bound retains the deterministic, point-value, univariate, fixed-$\alpha$ oracle restrictions and its function class. R121 supplies a constructive original-model acquisition interface and its actual paid queries, not a broader lower-bound proof. The direct original-formula online control is retained in all economic comparisons; excluding it on the basis of the narrow oracle theorem would give an unfair computational claim.

**Retained R121 remaining scientific requirement.** Randomized, multivariate, nonuniform and formula-oracle lower bounds remain separate open statements. The actual original-model information bill does not itself establish minimax optimality.

**Retained R121 execution and source route.** Theorem S.4, Proposition S.5, and the acquisition and complete-frontier rows. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** High-level references to the retained lower bound state deterministic, point-value, univariate, fixed-alpha scope. The new local theorem separately spells out exact supplied support information and constant-curvature witnesses. The implementation's finite probes and extra original-gradient enclosure are identified where they change sharpness.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** The qualification must remain adjacent whenever the inherited result is summarized, including in later abstracts and discussion edits.

### 0.23 Minor 3. Tail dependence

**Original advisory request.**

> 3. Discuss the opposite tail-level behavior of the lower and upper constants as 'alpha' approaches one.

**Completed R120 author response.** The achieved R119 order $KR(1-\alpha)N^{-2}$ is preserved. R120's fixed-tail comparison intersects widths bounded by $2KR(1-\alpha)/n^2$ and $ER(1-\alpha)/n$, with the original smooth coefficient supplied separately. The action modulus includes its explicit $\alpha/(1-\alpha)$ dependence; this prevents a small tail mass alone from implying a cheap original-model computation near $\alpha = 1$. The full-support $\alpha = 0$ boundary is separate and the normalized-tail statement

excludes $\alpha = 1$. Actual new observations use only the predeclared $1/2$ and $3/4$ risk levels and retain every paid tail/terminal operation.

The completed evidence for this item is in Table AE.II, Table AE.I; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** The original-risk regularity and probe propositions retain explicit dependence on the tail probability and capacity constants, while the inherited fixed-tail lower bound keeps its own tail-dependent constant and function class. The executed comparison reports both $\alpha = 1/2$ and $\alpha = 3/4$ separately throughout the acquisition, regret, loss and cost tables; the 28 paired sample mean gains are negative across both risk levels. No data-dependent risk level has been chosen to improve the claim.

**Retained R121 remaining scientific requirement.** The two fixed risk levels do not provide empirical evidence about the $\alpha \uparrow 1$ limit. Opposite movements of constants from different bounds or function classes cannot be combined into a uniform sharp-rate claim without the corresponding assumptions and proof.

**Retained R121 execution and source route.** Theorem S.4, Proposition S.5, and Table X.I. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The retained continuous-risk bounds have distinct dependence on the risk level and opposite tail-limit behavior. The original smoothness constant $L_\alpha$ is displayed in the exact primitive derivation, including its dependence on the ratio of $\alpha$ to $1 - \alpha$. These constants affect the support-derived action uncertainty and candidate width, so numerical conclusions remain specific to the two predeclared tail levels.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** No tail-uniform lower/upper complexity equivalence or extreme-tail numerical performance is established by the fixed-alpha theorem or this two-level study.

## 0.24 Minor 4. Acquisition restrictions

**Original advisory request.**

> 4. The readout coefficient bounds are acquisition restrictions, not characterizations of the best approximation; report sensitivity.

**Completed R120 author response.** The actual actor artifacts and acquisition table preserve all prescribed feature counts, coefficient signs/bounds, rounding and ties. The constrained rank-six readouts are compared with the larger fixed polynomial and grid controls under the same 256 original-$V_1$ label sites per risk. ReLU and ordinary hinge have identical retained parameters and actions, with the full shared-fit charge assigned to each standalone deployment. These are acquired-model comparisons under known restrictions. They are not best-approximation results for unrestricted representation classes, and the experiment supplies no post-outcome coefficient or architecture sensitivity menu.

The completed evidence for this item is in Table AE.I, Table AE.II; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** The actual fitting problem fixes a coefficient box of 128 and 30-bit coefficient storage; its exact audit certifies this finite feasible class. All 28 MARGINAL primal/dual gap upper bounds are at most approximately $2.58 \times 10^{-8}$, and VALUE least-squares gaps are separately reported for the VALUE actors. The two acquired budgets change information and some feature counts, but the coefficient box is unchanged. We therefore report the restriction and its exact solved-problem gap without interpreting it as the best unrestricted approximation or as a coefficient-box sensitivity experiment.

**Retained R121 remaining scientific requirement.** Sensitivity to the coefficient bound has not been executed in R121. Changing that bound, features or selection after seeing these outcomes would require a new declared experiment; the present two-budget comparison cannot answer the omitted box-sensitivity request.

**Retained R121 execution and source route.** Proposition S.6, Table AA.III, and Table AA.VI. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The finite readout boxes remain acquisition and optimization constraints attached to the actual historical models. Their exact labels, coefficients, feature identities and independent finite-box audits are preserved. The action safeguard uses original paid support and curvature, not an assertion that the bounded readout class characterizes best approximation. The two inherited budgets are retained without retuning their boxes after new outcomes.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** R122 adds no new coefficient-bound sensitivity experiment or globally optimal representation claim. Broader box sensitivity requires a separately frozen acquisition and full downstream comparison.

## 0.25 Minor 5. Residual span and action gap

**Original advisory request.**

> 5. Distinguish validated residual span from Bellman action gap in terminology and tables.

**Completed R120 author response.** Every primary group now has separate original-policy bounds, independent original-optimum bounds, mean regret, RMS action-error upper bound and own-incumbent gain, alongside production contrast and certificate counters. The independent validator supplies 55,104 retained all-action support rows and all 4,096 policy comparisons. For example, the ReLU P gain is economically positive against its incumbent while its mean difference from L is exactly zero. A narrower value interval, smaller fit residual or lower regret ceiling is therefore never counted as an observed policy-value decrease by itself. The 726 versus 3,370 containment directions also retain their different independent-verification implications.

The completed evidence for this item is in Table AE.I; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** The result tables distinguish finite-node marginal residual, finite-box optimization gap, production qualification, independently bounded original-policy regret and action-error upper bounds. The distinction has empirical content: all 28 node residuals decrease while all 28 original sample mean gains are negative. All 3,712 policies pass production qualification, yet that does not establish a better learned decision than VALUE or a neural adoption advantage. Production and independent interval overlap is described only as compatibility; it does not independently certify the narrower production endpoints.

**Retained R121 remaining scientific requirement.** A node residual or fitting gap is not an off-design Bellman action gap. The proved feature-cover and KKT implications require their stated hypotheses; the fitted global cover was not numerically certified, and action-error upper bounds are not exact observed action errors.

**Retained R121 execution and source route.** Theorem S.7, Proposition S.8, Table X.I, and Table AA.I. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The response separates critic fitting loss, validated residual span, support gap $e = U - W$, original derivative uncertainty, robust action gain, complete-policy regret and independent evaluation width. The main safeguard requires original-policy improvement, regret and width together. A support slope is explicitly not the original derivative, particularly at a capacity boundary, and a raw independent regret enclosure is not silently replaced by the proof-backed production cap.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** An unresolved derivative sign or interval-width test remains an unresolved action decision. Better internal approximation diagnostics do not become a Bellman action or economic-loss certificate by terminology.

### 0.26 Minor 6. Rejections and unsuccessful work

**Original advisory request.**

6. Report rejection changes beside every span improvement.

**Completed R120 author response.** Every attempted candidate and resolution, rejection, exhausted ladder, unchanged action and fallback is retained in the actual traces and complete tables. P's 2,257 refusals and D's 633 refusals appear beside accepted-step counts, as do 785 and 194 exhausted ladders. Mixed-bound use is reported at all 9,016 paid levels: 5,961 active upper bounds and two necessary accepted records for one economic candidate. The transferred incumbent bound is inactive in all 2,290 density pairs. Its mathematical validity is not converted into an unobserved reduction of already purchased tail endpoints, candidate-upper work or total time. Every unresolved and adverse comparison remains included.

The completed evidence for this item is in Table AH.III; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** Each of the 28 loss-contrast rows reports the node-residual change beside the first-rejection change and the independently bounded original gain. Although every node residual improves, rejection counts rise in 18 paired treatments, fall in two and are unchanged in eight; the aggregate is 294 MARGINAL rejections versus 60 VALUE rejections. All 354 rejected first proposals are kept, independently evaluated and charged together with fallback. The raw-proposal table and delivered-policy table distinguish a poor learned suggestion from the corrected policy ultimately returned.

**Retained R121 remaining scientific requirement.** A smaller residual is not evidence of fewer rejections or safer raw actions. The 28 treatments reuse the same states and include identical-feature controls, so these counts are descriptive treatment totals rather than 28 independent draws.

**Retained R121 execution and source route.** Table X.I, Table AA.II, and the complete qualification-attempt rows. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The complete saved tables report original actor rejection and fallback, changed actions, all algebraic candidates and failed certificate tests, paid refinement starts/completions, exact fallback and enhancement failures beside economic gains and widths. The prior increase in R121 initial actor rejection remains visible beside its marginal-fit improvement. Accepted-only independent confirmation uses the accepted-action denominator and excludes unchanged-action exact-zero comparisons.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** No method may be selected or declared improved solely because a fitting or span diagnostic falls while its original rejection, cost or delivered-policy outcomes deteriorate.

### 0.27 Minor 7. Observed engineering ranges

**Original advisory request.**

> 7. Label the acquisition bracket as an observed engineering range, not a confidence interval.

**Completed R120 author response.** The three process-specific measurements, all 192 cohort clocks and their medians and ranges are labeled descriptive engineering observations. All 6,144 comparisons of actions, intervals and query counts match, so the reported conditional pairing of timing and economic outcomes is actually checked. These clocks remain distinct from outward rational value enclosures and from the explicitly proved simultaneous 95% population family of 48 groups and 192 events. Timing replicates do not increase its 64 independent initial-state units. The inherited seven R119 clocks retain their separate episode and do not enlarge the current sample.

The completed evidence for this item is in Table AE.II, Table AE.III; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** The acquisition ledger records the two actual processes and the component allocation used for deployment comparisons; it does not assign sampling coverage to the observed engineering costs. The five fresh service clocks are all reported, with 40 faster and 100 slower MARGINAL cohort comparisons, or 38 and 102 after the allocated acquisition and audit bill. Raw clock observations, complete enclosing processes and calculated reuse costs have separate fields. The independent original-policy validation cost remains visible in the research ledger.

**Retained R121 remaining scientific requirement.** Observed minima, maxima, medians and component allocations are not runtime confidence intervals. The fixed 95% family concerns economic state sampling, not runtime variation; timing repeats do not enlarge its sample.

**Retained R121 execution and source route.** Table AA.V, Table AA.IV, and the acquisition/research process records. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The earlier acquisition bracket remains an observed engineering range, not a confidence interval. R122 carries the actual historical acquisition allocation and separately labels each current descriptive process clock, declared amortization scenario and partial research-cost lower bound. Only the fixed state-level inferential families receive the stated simultaneous coverage.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** Clock variation, incomplete publication coverage and the interrupted parent are not converted into statistical confidence intervals or unobserved exact totals.

### 0.28 Minor 8. Adoption over the conventional frontier

**Original advisory request.**

8. State visibly that there are zero adoption regions over quadratic least squares.

**Completed R120 author response.** The old zero-adoption-over-quadratic result and the R119 conventional frontier remain explicit. In the new fixed comparison, positive gains against learned incumbents coexist with an online advantage against all 14 learned completed means and exact ReLU–hinge identity. All 32 ReLU intersections against the predeclared full conventional menu, across four amortization volumes and three clocks, have empty sufficient-price sets and empty intersections of upper not-ruled-out sets. Among individual clock comparisons for ReLU, 12 of 96 upper regions remain nonempty; two of eight actual-primary workload upper regions are also nonempty. These are unresolved comparisons, retained separately from the three-clock intersections. The 17 nonempty sufficient menu regions across representations belong to completed conventional policies. The 112 separate actual-64-state price domains retain the actual workload and charges. These strict recorded-menu conditions and own-incumbent gains do not assert population adoption or rule on every possible deployment criterion.

The completed evidence for this item is in Table AH.VIII, Table AE.I; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** The original R97 zero certified adoption result over quadratic least squares and the exact R120 individual-clock qualifications remain explicitly preserved above. The completed R121 full-menu result is now stronger for its own declared scenarios: zero of 32 primary ReLU price cells, zero of 160 individual timing cells and zero of 32 all-five-clock intersections has a nonempty sufficient or upper not-ruled-out region. The predeclared full menu includes quadratic least squares and the other conventional alternatives at both budgets. Every pair constraint remains in the complete table, so the full-menu result is not silently relabeled as a separate pairwise claim against quadratic alone.

**Retained R121 remaining scientific requirement.** An empty full-menu region does not logically imply that every pairwise comparison is impossible, and the older unresolved R120 individual-clock regions remain unchanged. These R121 finite-cohort, measured-cost conditions are not population adoption or a universal impossibility theorem.

**Retained R121 execution and source route.** Table AA.VII and every exact pair constraint, menu and all-five-clock intersection. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The original zero-adoption statement over quadratic least squares remains attached to its own historical report and experiment. Current R122 conventional comparisons include within-budget quadratic/VALUE/control in all 348 signed

tests and the full same-risk conventional menu in every price scenario. Both sufficient and upper not-ruled-out domains are retained, including every empty region, and current numerical conclusions are populated only from actual R122 saved tables.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** Neither an old zero count nor a new favorable single scenario may be substituted for the current full-menu and common-clock result. A conditional sample domain does not identify a population neural adoption advantage.

### 0.29 Minor 9. Verified approximation and HJB literature

**Original advisory request.**

> 9. Compare the certification contribution carefully with existing verified approximation and HJB-residual literatures.

**Completed R120 author response.** The main text locates the mixed integration interface relative to retained verified approximation, Bellman/HJB residual ideas, constrained safeguards and the source-owned common-risk identity. Complete proofs retain the original filtration, boundaries and one-sided future-policy allowance. Exact binomial inversion and bounded-mean concentration are attributed as classical ingredients; the new population conclusion is the specified transfer from actual certified gains. The stopped Brownian verification subcase retains its hypotheses and complete boundary argument. The present two-date validation supplies no proof or experimental substitute for a general coupled NDU/FBSDE, nonlinear recursive-preference application, temporal-self equilibrium or game.

The completed evidence for this item is in Table AE.III; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** The certification account retains the verified approximation, Bellman/HJB residual, recursive-utility and safeguarded-policy connections with their original hypotheses and attribution. R121 adds a complete original-risk marginal construction through constrained tails, a paid interval-label interface, a finite fitting witness and an explicit nonsmooth feature-range transfer. The independently implemented finite audit and original-policy evaluator have both executed, while the global fitted-feature cover remains unexecuted. Classical concentration and binomial inversion ingredients do not become new neural theorems because they support the fixed R121 evidence.

**Retained R121 remaining scientific requirement.** The preserved NDU/FBSDE, nonlinear preference, temporal-self and game links require their own well-posedness, filtration, boundary and equilibrium assumptions. The present two-date investment validation neither proves nor executes those different economic claims.

**Retained R121 execution and source route.** The complete new proofs, retained correctness/bridge sources, Table AA.III, and the independent original-policy receipts. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

**Current R122 mathematical, algorithmic and author action.** The supplement retains the verified approximation, continuous-time comparison, residual and stability foundations with their exact assumptions. The current main text also positions its calculation within inexact first-order oracle and smooth-convex interpolation work. Its specific claim is the conversion of a paid original Bellman minorant into constrained action information and a checked implemented-policy replacement. It does not claim the generic descent lemma, projected gradient method or residual certification as new. Theorem S.24 now gives the exact finite-horizon same-incumbent gain recursion for the original conditional risk operator. A nonnegative advantage certified at every admissible state propagates through a centered lower operator with nonnegative dual slack. Proposition S.25 gives its finite complete-menu minimum-occupation calculation, and Proposition S.26 states the discounted bounded-measurable fixed-point version. The proof separates physical from risk-distorted occupation and requires a valid lower bound on the entire dual infimum. It changes no current experiment and proves neither neural exclusivity nor a new general policy-iteration convergence rate.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** A proof-backed local action statement still requires its numerical premises to hold. It does not by itself validate a learned HJB/FBSDE residual or transfer economic objectives between continuous and discrete models.

### 0.30 Minor 10. Title and neural content

**Original advisory request.**

> 10. The title should not carry more neural-specific content than the theorem and evidence identify.

**Completed R120 author response.** The title and active main text retain Neural Bellman Operators as the study of learned original continuations, recursive operators and feasible actor/critic decisions. The new action-dependent comparison and safeguard are attributed to an eligible-method interface shared by the neural and conventional representations. Their original-policy gains are actually measured and independently bounded; conventional hinge remains identical to ReLU, and the strongest original-formula controls remain visible. This establishes a concrete shared numerical-methods result inside the NBO problem while leaving equal-information neural advantage and action-aware learning as explicit ongoing research requirements.

The completed evidence for this item is in Table AE.I; the exact row files and hashes are retained in the current response map.

**Completed R121 mathematical, algorithmic and empirical action.** Neural Bellman Operators remains the research identity: recursive continuation operators, their learned critics and feasible actors are studied under the original investment objective. The title and contribution distinguish the neural implementation from the representation-neutral theory. Current evidence is stated directly: the marginal objective is accurately solved at the finite nodes, yet every paired sample mean worsens, matched hinge actions equal ReLU, and no ReLU full-menu adoption region is obtained. These findings inform the next original-model algorithmic problem rather than redefining a shared hinge geometry as neural-specific success.

**Retained R121 remaining scientific requirement.** The title does not assert an established neural-exclusive economic or information advantage. Achieving such an advantage with equal information, feasible policies and complete costs remains part of the continuing NBO and Econometrica objective.

**Retained R121 execution and source route.** The revised main contribution statement, Table X.I, and Table AA.VII. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

# 1 Historical preservation, unresolved episodes and publication scope

All earlier original report text and author response routes remain unchanged in the compact inherited maps indexed by `provenance121/reports121/REPORT_SOURCE_INDEX121.json`. The 620 major routes, 68 explicit minor requests, 295 separately retained supplementary requirements, 31 embedded requests and 35 overlapping parent treatments total 1,049 authored route records. The overlapping parent treatments are not added as independent scientific issues. Exact source texts and their original commit identities remain authoritative.

R98 and R114/R115 retain their independent protocols, selected-model and sample barriers. A previously recorded not-started state is a dated observation, not a live process-status assertion. R111–R118 retain their separate model and information conventions, incomplete handoffs, failed transport and actual later reproductions where documented. In particular, R119's actual unchanged R117/R118 reproductions do not complete the independent R115 population study. R118's inherited R110 article and truncated transport do not constitute a newer completed author baseline. The retention map records these episode-specific boundaries: `CONTENT_RETENTION_MAP121.md`.

The original-theory and source registries distinguish a complete file or cross-reference census from full semantic reading and from fresh mathematical revalidation. Corrected historical claims remain visible with their corrigenda; their original source is not silently edited. The report reading-coverage receipt records the actual reader and scope of each full report reading rather than inferring reading from a hash.

R120 source CI run 39240515942 actually completed successfully at its bound source and produced artifact 12181820272. Its independently read artifact and exact four-PDF comparison are retained as R120 evidence. They are not R121 CI. The final R121 source, PDF and publication declarations will be separate later records bound to their actual outputs; a queued run, a successful archive copy or a bounded reporting replay will not be described as a complete scientific rerun.

**Current R122 mathematical, algorithmic and author action.** The title remains Neural Bellman Operators to preserve the original research program. The abstract and main text identify the

acquired continuation/actor construction, the shared original-information safeguard and the strong conventional controls. They distinguish a useful component of neural Bellman computation from a neural-exclusive information, approximation or economic advantage.

**Current R122 evidence route.** The exact current source and saved-stage bindings for this item are in `provenance122/R97_REQUIREMENTS122.json`. The actual execution scope at the beginning of this response applies to this item; no historical outcome is substituted for a missing current stage.

**Current remaining scientific requirement.** The title does not imply that the present theorem proves neural superiority. Such a claim remains subject to matched information, policy and complete-resource evidence.